



HW

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Ch4_Q13(efghij)

- (e) Repeat (d) using LDA.
- (f) Repeat (d) using QDA.
- (g) Repeat (d) using KNN with $K = 1$.
- (h) Repeat (d) using naive Bayes.
- (i) Which of these methods appears to provide the best results on this data?
- (j) Experiment with different combinations of predictors, including possible transformations and interactions, for each of the methods. Report the variables, method, and associated confusion matrix that appears to provide the best results on the held out data. Note that you should also experiment with values for K in the KNN classifier.



Ch4_Q14

14. In this problem, you will develop a model to predict whether a given car gets high or low gas mileage based on the `Auto` data set.

- (a) Create a binary variable, `mpg01`, that contains a 1 if `mpg` contains a value above its median, and a 0 if `mpg` contains a value below its median. You can compute the median using the `median()` method of the data frame. Note you may find it helpful to add a column `mpg01` to the data frame by assignment. Assuming you have stored the data frame as `Auto`, this can be done as follows:

```
Auto['mpg01'] = mpg01
```

- (b) Explore the data graphically in order to investigate the association between `mpg01` and the other features. Which of the other features seem most likely to be useful in predicting `mpg01`? Scatterplots and boxplots may be useful tools to answer this question. Describe your findings.
- (c) Split the data into a training set and a test set.
- (d) Perform LDA on the training data in order to predict `mpg01` using the variables that seemed most associated with `mpg01` in (b). What is the test error of the model obtained?



- (e) Perform QDA on the training data in order to predict **mpg01** using the variables that seemed most associated with **mpg01** in (b). What is the test error of the model obtained?
- (f) Perform logistic regression on the training data in order to predict **mpg01** using the variables that seemed most associated with **mpg01** in (b). What is the test error of the model obtained?
- (g) Perform naive Bayes on the training data in order to predict **mpg01** using the variables that seemed most associated with **mpg01** in (b). What is the test error of the model obtained?
- (h) Perform KNN on the training data, with several values of K , in order to predict **mpg01**. Use only the variables that seemed most associated with **mpg01** in (b). What test errors do you obtain? Which value of K seems to perform the best on this data set?



Ch4_Q15

15. This problem involves writing functions.

- (a) Write a function, `Power()`, that prints out the result of raising 2 to the 3rd power. In other words, your function should compute 2^3 and print out the results.

Hint: Recall that `x**a` raises `x` to the power `a`. Use the `print()` function to display the result.

- (b) Create a new function, `Power2()`, that allows you to pass *any* two numbers, `x` and `a`, and prints out the value of `x**a`. You can do this by beginning your function with the line

```
def Power2(x, a):
```

You should be able to call your function by entering, for instance,

```
Power2(3, 8)
```

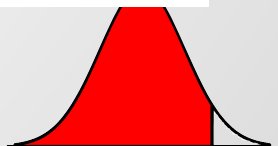
on the command line. This should output the value of 3^8 , namely, 6,561.

- (c) Using the `Power2()` function that you just wrote, compute 10^3 , 8^{17} , and 131^3 .

- (d) Now create a new function, `Power3()`, that actually *returns* the result `x**a` as a `Python` object, rather than simply printing it to the screen. That is, if you store the value `x**a` in an object called `result` within your function, then you can simply `return` this result, using the following line: `return`

```
    return result
```

Note that the line above should be the last line in your function, and it should be indented 4 spaces.



- (e) Now using the `Power3()` function, create a plot of $f(x) = x^2$. The x -axis should display a range of integers from 1 to 10, and the y -axis should display x^2 . Label the axes appropriately, and use an appropriate title for the figure. Consider displaying either the x -axis, the y -axis, or both on the log-scale. You can do this by using the `ax.set_xscale()` and `ax.set_yscale()` methods of the axes you are plotting to.

```
.set_xscale()  
.set_yscale()
```

- (f) Create a function, `PlotPower()`, that allows you to create a plot of x against $x**a$ for a fixed a and a sequence of values of x . For instance, if you call

```
PlotPower(np.arange(1, 11), 3)
```

then a plot should be created with an x -axis taking on values 1, 2, ..., 10, and a y -axis taking on values $1^3, 2^3, \dots, 10^3$.





TEN Template

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